

National University of Computer and Emerging Sciences, Lahore Campus

	Course Name	Software Design & Analysis	Course Code	CS3002
	Name		Exam Type	Quiz 5.v2
	Roll.No		Total Marks	/10
	Section:	5J	Weight	

Note : HAPPY LAST QUIZ ! , BEST OF LUCK FOR FINALS

Q1 : Improve the following code using an appropriate design pattern and give the solution code:

```
struct Pair {
    Student* x;
    Student* y;
};
void foo() {
    Pair pr1 = Student::getPair();
}
void bar() {
    Pair pr2 = Student::getPair();
}
```

When function **foo()** calls **getPair()**, it gets a pair object containing pointers to two objects of the Student class.

Afterwards, when function **bar()** calls **getPair()**, it gets a pair object containing the **same** two Student objects again!

Solution:

Student Class (Singleton)

```
class Student {
private:
    static Student* s1;
    static Student* s2;

    // Private constructor
    Student() {}

public:
    static Pair getPair() {
```

```
    if (s1 == nullptr) {  
        s1 = new Student();  
        s2 = new Student();  
    }  
    Pair p;  
    p.x = s1;  
    p.y = s2;  
    return p;  
}  
};
```

Define Static Members

```
Student* Student::s1 = nullptr;  
Student* Student::s2 = nullptr;
```

Now foo() and bar() both return SAME two Student objects

```
void foo() {  
    Pair pr1 = Student::getPair(); // same two objects  
}
```

```
void bar() {  
    Pair pr2 = Student::getPair(); // same two objects  
}
```